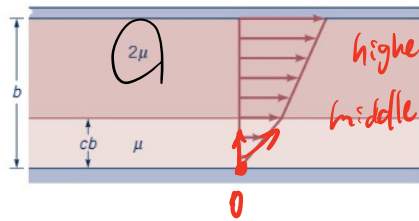


Howard Kuo ChemE 0905

ME106: Assignment 1

1. Two flat plates are oriented parallel above a fixed lower plate as shown in the figure below. The top plate, located a distance b above the fixed plate, is pulled along with speed V . The other thin plate is located a distance cb , where $0 < c < 1$, above the fixed plate. This plate moves with speed V_1 , which is determined by the viscous shear forces imposed on it by the fluids on its top and bottom. The fluid on the top is twice as viscous as that on the bottom. Plot the ratio $\frac{V_1}{V}$ as a function of c for $0 < c < 1$.



lowest fluid layer: $\mu = \frac{\tau}{\frac{du}{dy}} = \frac{\tau}{\frac{V - 0}{b - cb}}$

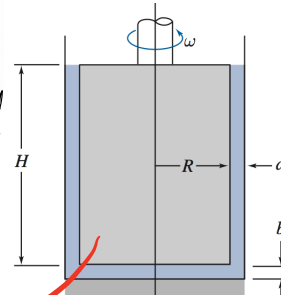
Upper layer: $2\mu = \frac{\tau}{\frac{du}{dy}} = \frac{\tau}{\frac{0 - V_1}{cb - 0}}$

Munson, 8th edition

2. Now consider a more practical shear flow. Shown below is a cross-section of a viscometer made of concentric cylinders. The inner cylinder spins with angular velocity ω and the outer cylinder remains stationary. The shearing of the fluid exerts a torque on the sides and bottom of the cylinder. Suppose the inner cylinder has height $H = 15$ cm and radius $R = 6$ cm; the gaps are $a = 1$ mm and $b = 1$ cm; and $\omega = 1$ s⁻¹. If the total torque on the cylinder is measured to be 0.1 N m, what is the viscosity of the fluid? (It is recommended to solve the problem in terms of the variables and then plug in numeric values at the end.)

- Assume a linear radial velocity profile.

- Use the simple shear relation $\tau = \mu \frac{du}{dy}$.
- shear stress: force on liquid tangent surface



velocity $\propto \omega$
pressure speed.

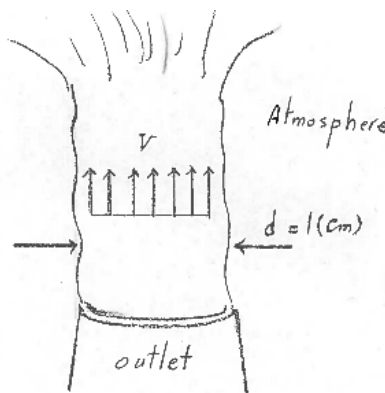
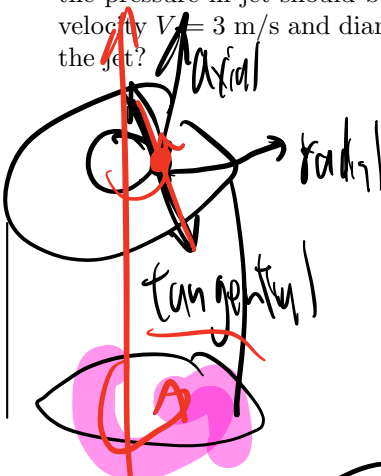
angular frequency
 $\omega = \frac{d\theta}{dt}$

$I = I_{\text{side}} + I_{\text{bottom}}$
Torque formula:

Torque: torque = force applied $\times$ distance.

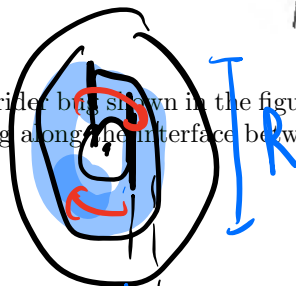
3. Water is pulled up trees in part due to the capillary effect. Assume that the capillaries in a tree are 50 microns in diameter. How far will water be carried up due to the capillary effect assuming the water is at 60°F and has a contact angle of 15°? (See inside cover of textbook for properties of water.)

4. A free jet is a stream of fluid exhausted to the atmosphere (such as sketched below). The pressure of the fluid inside the free jet is usually assumed to be atmospheric pressure. However, if we do not neglect surface tension, the pressure in jet should be slightly higher than atmospheric pressure. Thus, consider a free jet of water with velocity $V = 3$ m/s and diameter 1 cm at $T = 20^\circ\text{C}$. How much does surface tension increase the pressure inside the jet?



surface of jet
du Tangent velocity
Newton law of viscosity
 $\mu = \eta \frac{du}{dy}$

5. (a) The water strider bug shown in the figure below is supported on the surface of a pond ($T = 20^\circ\text{C}$) by surface tension acting along the interface between the water and the bug's legs. Determine the minimum length of

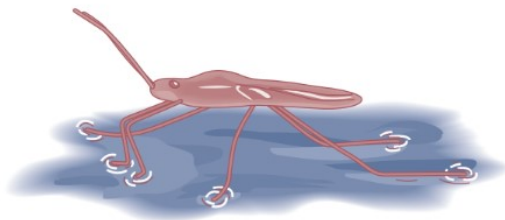


$\frac{du}{dy} = \frac{R \cdot \omega (\text{cm} \cdot \text{s}^{-1})}{a}$

γ (N/m)
 (1s) inner surface (0s) outer surface
 TS
 Liquid at inner face has speed max

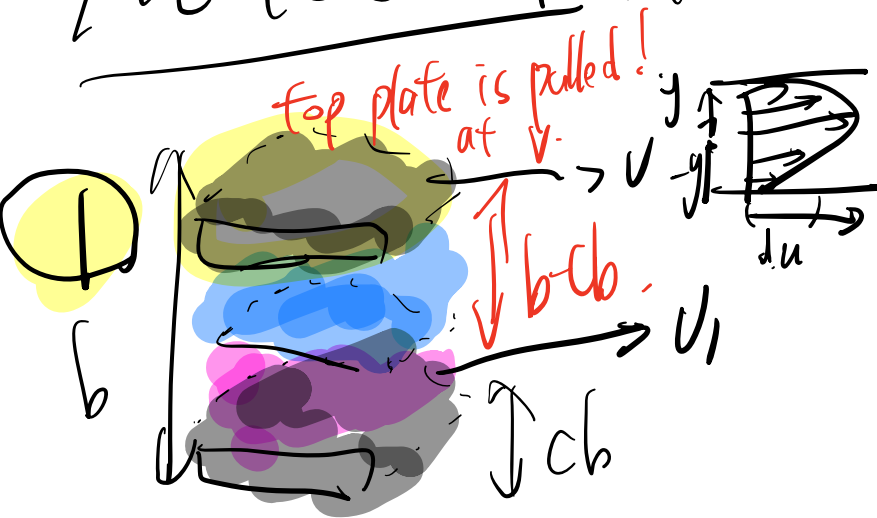
this interface needed to support the bug. Assume the bug weighs 10^{-4} N and the surface tension force acts vertically upwards.

- (b) Repeat part (a) if surface tension were to support a person weighing 750 N.



Munson, 8th edition

ME106 W1 Howard Kuo



Discussion:

Physical meaning of

equal shear stress ??

Newtonian's Law

$$\tau = \mu \cdot \frac{dy}{dx} \quad \tau_{\text{top}} = 2\mu \cdot \frac{(V - V_1)}{(b - c_b)}, \quad \tau_{\text{bot}} = \mu \cdot \frac{(V_1 - 0)}{c_b}$$

for middle plate: $\tau_{\text{top}} = \tau_{\text{bot}} \rightarrow 2\mu \frac{V - V_1}{b(1 - c)} = \mu \frac{V_1}{c_b}$

key character: for middle plate move in constant velocity V_1 :

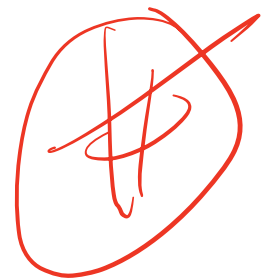
$$2c(V - V_1) = V_1(1 - c)$$

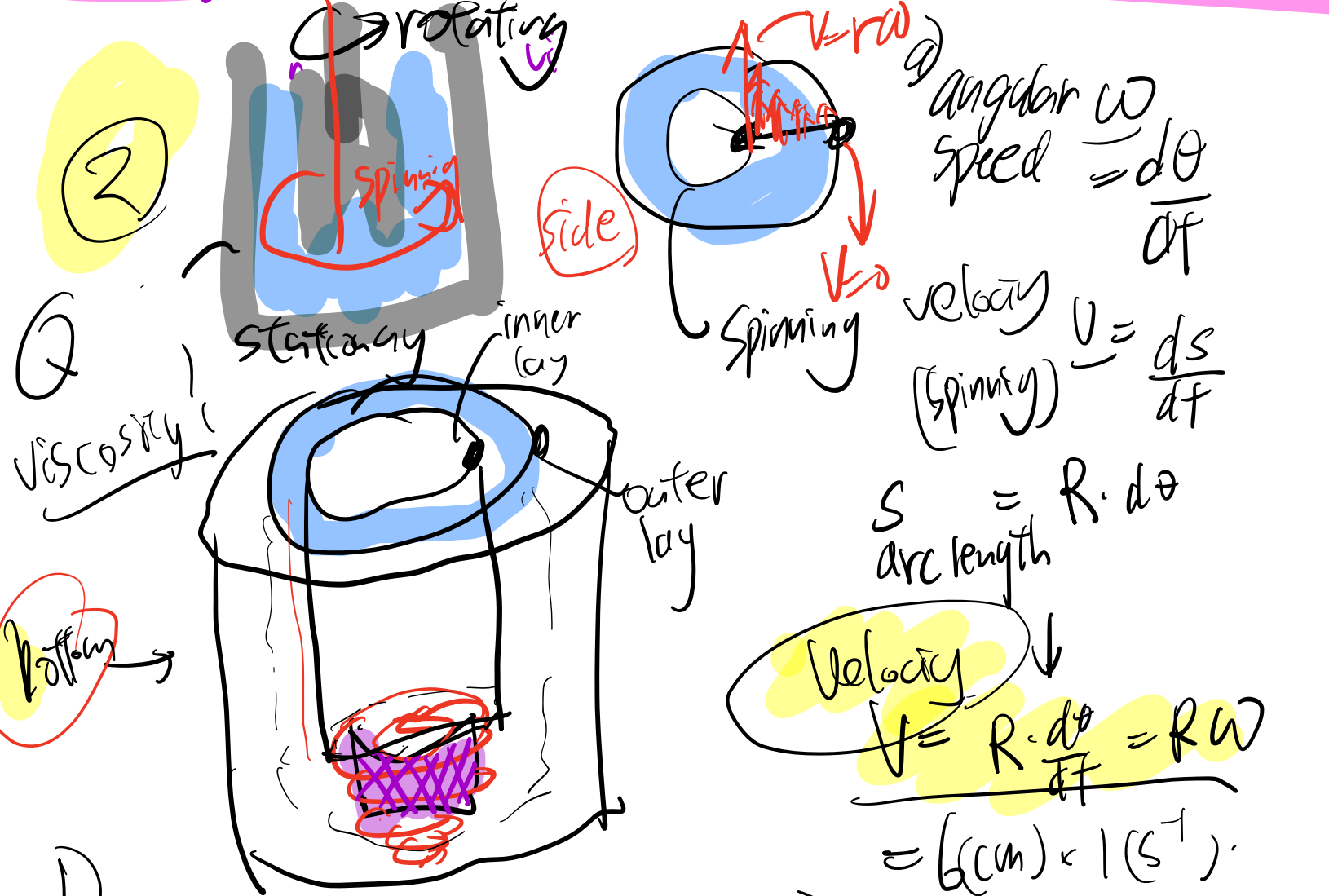
Shear stress balance = same
Horizontal balance

$$2cV = V_1(1 - c) + V_1(c)$$

$$\rightarrow \frac{V_1}{V} = \frac{2c}{(1 - c + 2c)} = \frac{2c}{c + 1}$$

* viscous
Analyze shear stress of the middle thin plate acting on





b) Shear stress of side

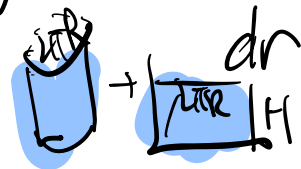
$$\tau_{\text{side}} = \mu \frac{du}{dy} = \mu \frac{0 - R\omega}{a}$$

$$\tau_{\text{bottom}} = \mu \frac{du}{dy} = \mu \frac{0 - \omega \cdot r}{b}$$

Consider circular ring



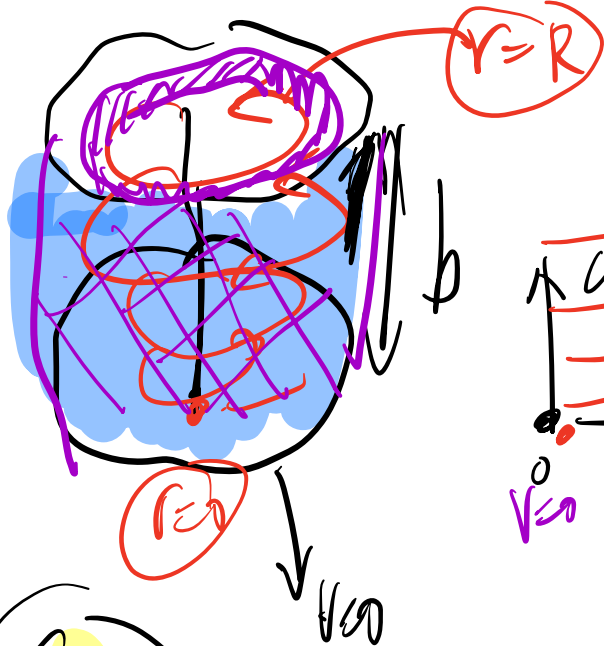
for radius r .



$$F_{\text{side}} = \mu \cdot \frac{R\omega}{a} \cdot 2\pi \cdot H \cdot dr$$

$$= -\frac{\mu \cdot \omega \cdot H}{a} \cdot R^2 \cdot 2\pi$$

r as rotation axis,



Area = $2\pi r \cdot dr$ (from 0 to R)

$v = \omega r$

$$F_{\text{bottom}} = \frac{\mu r \omega}{b} \cdot 2\pi r dr = 2\pi \frac{\mu r \omega r^3}{b}$$

d) torque = arm length $\times$ Force

$$dT = r \cdot dF = \int_0^R \frac{2\pi \mu \omega H r^2}{a} dr = \frac{2\pi R^3 \mu H \omega}{a}$$

T_{side}

$$dT_{\text{bottom}} = \int_0^R \frac{2\pi \mu r \omega r^3}{b} dr = \frac{\pi \mu \omega R^4}{2b}$$

$$T_{\text{total}} = T_{\text{side}} + T_{\text{bottom}} = (\mu \omega \pi) \left[\frac{R^3 H}{a} + \frac{R^4}{2b} \right] = 0.1$$

$$\mu = \frac{0.1}{\left[\frac{(0.06)^3}{0.001 \text{ mm}^3} \cdot 0.15 \text{ m} \right] + \left[\frac{(0.06)^4}{2 \cdot 0.01} \right] \times 1 \text{ xT}}$$

$$\rightarrow \mu \approx 0.486 \text{ Pa}\cdot\text{s}$$



Density

$$\rho = 62.37 \frac{\text{lb}}{\text{ft}^3} \quad \text{or} \quad \left(\frac{2.9}{\text{g}} \right) \frac{1}{\text{cm}^3}$$

$$\rho = 62.37 \frac{\text{lb}}{\text{ft}^3} = 5.03 \cdot 10^{-3} \frac{\text{lb}}{\text{ft}^3}$$

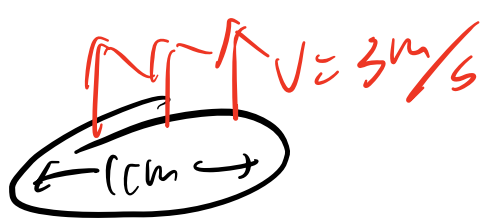
$$h = \frac{2 \cdot \rho \cdot R \cdot \cos \theta}{\rho \cdot R} = \frac{2 (5.03 \cdot 10^{-3}) \cdot \cos(15)}{62.37 - 82 \cdot 10^{-5}} \frac{\text{lb}}{\text{ft}}$$

$$R = 25 \times 10^{-6} \text{ m} = 25 \times 10^{-6} \times 3.28 \frac{\text{m}}{\text{ft}} = 8.2 \cdot 10^{-5} \text{ ft}$$

$$= 1.9 \text{ ft} = 579 \text{ mm}$$

④ Free jet 自由射流

Surface tension: force on unit length



Young-Laplace Equation

$$\Delta P = -\gamma \nabla \cdot \hat{n}$$

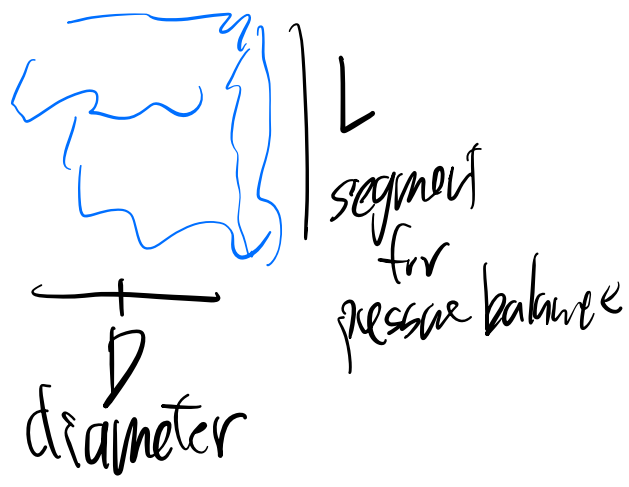
pressure difference in interface γ surface tension F/length $\nabla \cdot \hat{n}$ curvature (perpendicular to interface) unit normal vector

$P_{inside} \rightarrow P_{atm}$
to maintain cylinder



Projected area

$$P_{atm} = \frac{\text{Length} \cdot D}{A = D \cdot L}$$



$$F = P_{in} - P_{atm} \cdot D \cdot L$$

(Net pressure trying to separate the two halves)

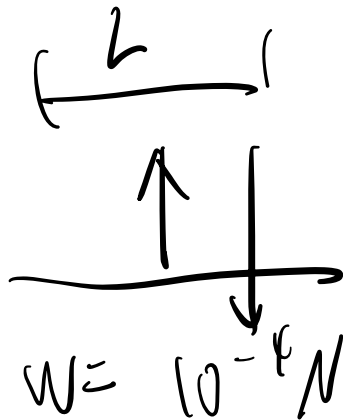
$$\sigma = \frac{F}{L}, F = \sigma \cdot L, F_b = (2) \cdot \sigma \cdot L$$

$$F_p = F_b \Rightarrow \Delta P \cdot DL = 2 \cdot 6 \cdot L \quad \sigma_{(H_2O, 20^\circ C)} = 7.28 \cdot 10^{-2} \text{ N/m}$$

$$\rightarrow \Delta P = \frac{(2) \sigma}{D} = 2 \cdot \frac{7.28 \cdot 10^{-2} \text{ N/m}}{1 \cdot 10^{-2} \text{ m}}$$

$$= 14.56 \frac{\text{N}}{\text{m}^2} = 14.56 \text{ Pa} \quad \#$$

5



$$\sigma_{(H_2O, 20^\circ C)} = 7.28 \cdot 10^{-2} \text{ N/m}$$

$$d) \sigma \cdot L = 10^{-4}, L = \frac{10^{-4}}{7.28 \cdot 10^{-2}}$$

$$= 0.00137 \text{ m}$$

$$b) \frac{750 \text{ N}}{7.28 \cdot 10^{-4}} = 10.302 \cdot 10^3 \text{ m}$$

$$\approx 10.3 \text{ km} \quad \#$$

$$L \approx 1.37 \text{ mm} \quad \#$$

Fundamental of fluid mechanics

Mausen et. al

TIA: Mon 1-3PM

CEM Hesse Hall
Lounge

Shawn

Tue
1-3PM

Readers

Missed Discussion Friday!

D Fundamental:

Spec. f.

flow
with
piper

Speed V ,

Pressure P ,

35 psi lb/in^2
pound force
square
inch

water from
fire hose:
 10^2 m/s

↓
Energy, Pressure

⌀ diameter

Fluid
Define

a substance that deforms
continuously when acted by a
shearing stress at any magnitude

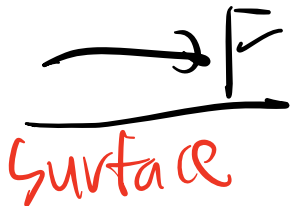
(Toothpaste, tar, slurries ...)

Either gas / liquid

Maintains Volume
of its container

Shear stress

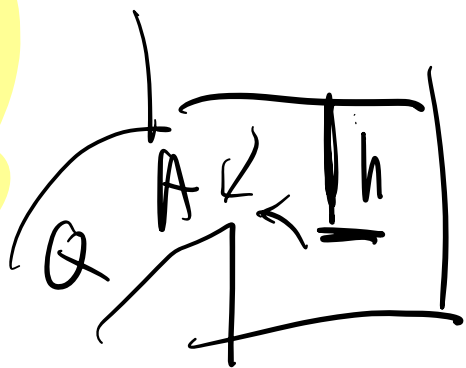
created when tangential force act on surface



act parallel to liquid layer

Google

Eq. 1



$$Q: \text{flow} = 0.61 A \sqrt{2gh}$$

= Volume / time = Liters / ~~cup~~ Cube

Size: h

$$\frac{l^3}{\text{time}} = (0.6) \cdot l^2 \cdot \sqrt{2 \cdot \frac{l}{\text{time}^2}} \cdot l$$

= meters / time² = l / time²

g
gravity acceleration

$$\frac{l^3}{\text{time}} = (0.6) \cdot l^2 \cdot \frac{l}{\text{time}} \cdot (?)$$

density

= ρ . g → specific weight

= $\frac{\text{mass}}{\text{volume}} \times \text{gravity}$

mass = slug = 14.6 kg

slug $\times$ ft/s² = lb → m · a = F

homogeneous

British Gravity

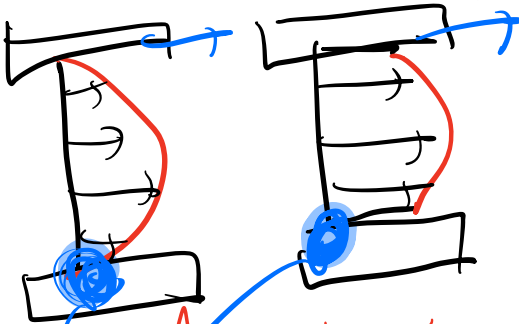
Viscosity

density, specific weight
(weight/volume).

Real fluid stick to solid boundaries although

maybe moving

Top plate moves in speed U



No Slip:

The fluid velocity component tangent to the solid surface is equal to the surface

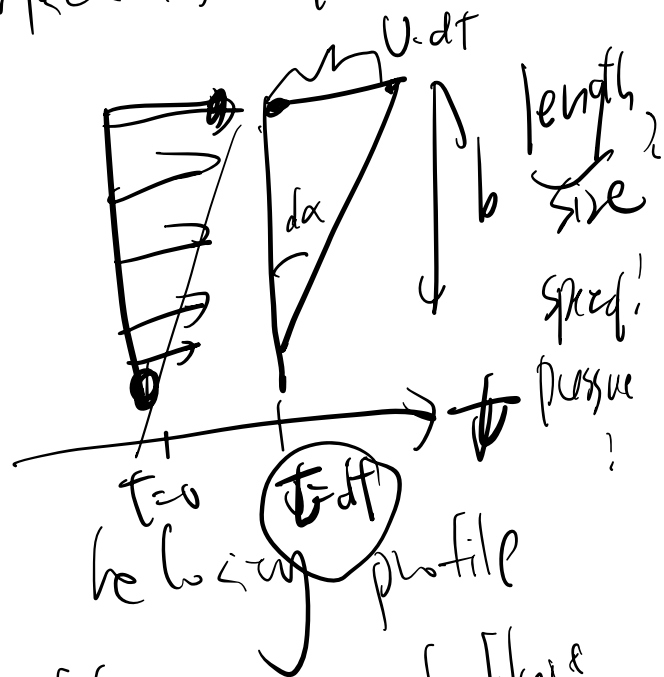
Boundary condition

The first fluid molecule sticks! (No slip)

$$\tan(\alpha) = \frac{\lim}{\cos} = \frac{U \cdot dt}{b}$$

$$= \frac{du \cdot dt}{dy} \quad \text{When angle is small} \quad \tan \alpha \approx \alpha \approx (d\alpha)$$

$$d\alpha = \frac{du \cdot dt}{dy}, \quad \frac{d\alpha}{dt} \text{ (change of angle)} = \frac{du}{dy} \rightarrow$$



[to ~ speed of fluid changes across a cross section path. Unit?

Newton Law of viscosity The shear stress, on a surface

tangent to the flow direction is proportional to the rate of change shear strain $\frac{d\alpha}{dt}$

$$\tau_{\text{wall}} \propto \frac{d\alpha}{dt} \rightarrow \tau \propto \frac{du}{dy}$$

shear stress

liquid $\frac{dN}{d\text{temp}} < 0$ gas > 0
 Heat $\uparrow$ viscosity $\downarrow$
 shear stress $\downarrow$

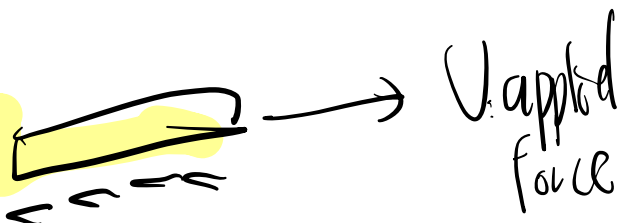
$$N = \tau \frac{du}{dy}$$

$$\tau = N \cdot \frac{du}{dy}$$

$N \rightarrow$ absolute viscosity

for newtonian fluid

blood is not newtonian



Shear stress

$$\tau = \frac{\text{applied force}}{\text{Area}}$$

Inviscid flow $\rightarrow (p=0)$
 viscosity $= 0$

$$\nu = \frac{\mu}{\rho}$$

Kinematic viscosity

ratio of absolute viscosity to fluid density

Ostwald-de Power Law for non-Newtonian liquid

$$\tau = k \left(\frac{du}{dy} \right)^n$$

$$k = \mu, n = 1$$

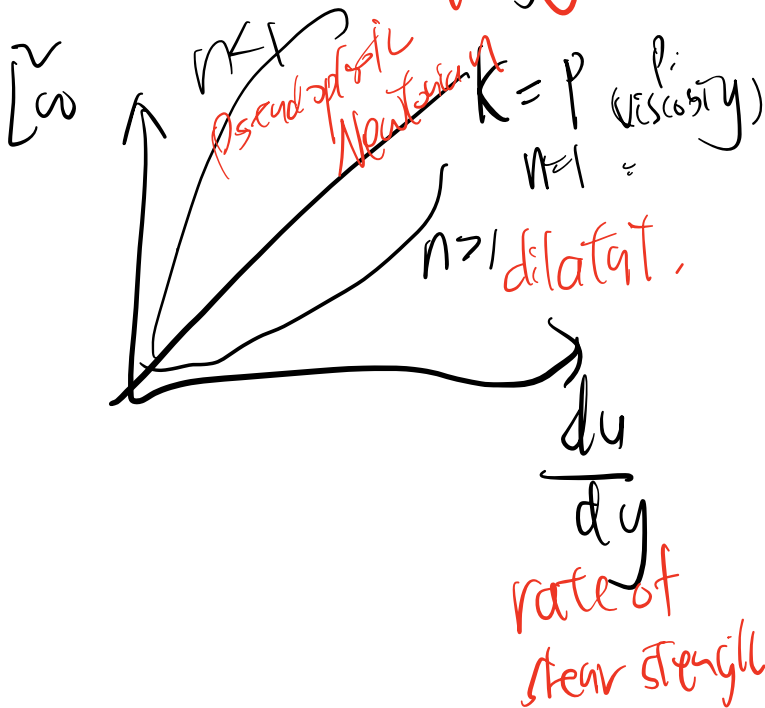
Shear stress $\tau = \frac{\text{force}}{\text{per unit area parallel to fixed surface}}$

Case by adjusting layer

Internal resistance to flow, or deformed by shear stress

$\mu = \frac{d\tau}{du}$

= shear stress unit



- Bulk Modulus $E_v = \frac{-dP}{dV/V} = \frac{dP}{d\rho/\rho}$

compression and expansion of gases

$$\frac{P}{\rho} = \text{constant}$$

- $E_v = P$ isothermal

$$c = \sqrt{\frac{E_v}{\rho}} = \sqrt{\frac{dP}{d\rho}}$$

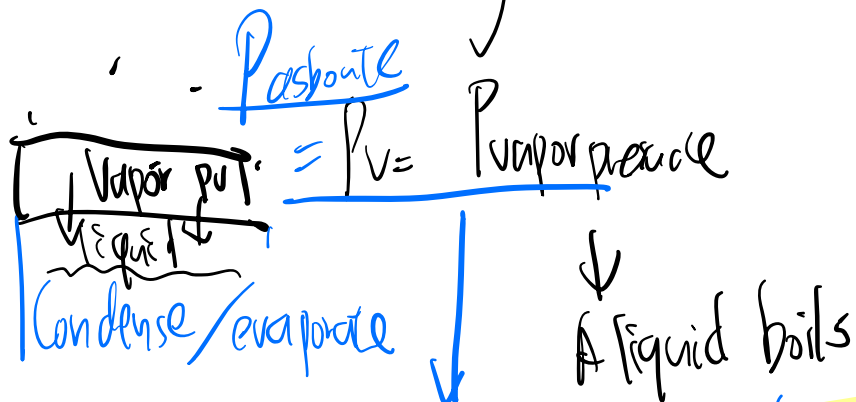
speed which fluid of disturbance sound propagate in

gas $Eu = KP$
 $C = \sqrt{\frac{KP}{\rho}}$
 $C = \sqrt{\frac{KRT}{M}}$
 is entropic ($ds = 0$)
 ideal gas law

Vapor pressure P_v : A liquid boils when pressure is reduced to the vapor pressure

in flowing liquid it is possible for the pressure in localized regions to reach vapor pressure

thereby causing cavitation



form of vapor bubble within fluid mass

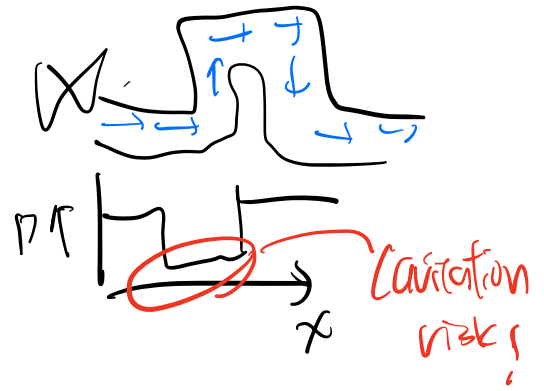
Equilibrium pressure exerted by vapor above it's liquid at Temp.

Q Boil vs. Cavitation?

eg. kitchen standard atmospheric pressure will boil
 $P_v = 14.7 \text{ PSI}$
 $T = 100^\circ\text{C}$

Same: liquid $\rightarrow$ vapor bubbles.
 diff: / increase in temp / vs / drop in pressure

Cavitation risk with elevated pipe:



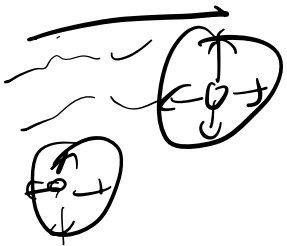
Surface
Tension

$$= \text{Force} / \text{length}$$

liquid gas interface
immiscible liquid

liquid surface

e.g. droplet of mercury form into sphere on smooth surface because cohesive force in the mercury surface hold molecule into compact,



e.g. 2

$$\sum F = ma$$

unit
conversion

$$F_{\text{floor}} - F_{\text{weight}} = ma$$
$$= mg$$

tank total mass

36 kg, rest

$$F = mg + ma$$

$$P_{\text{pressure}} = \rho RT$$

Ideal Gas Law $\xrightarrow{\text{Density}}$

Specific weight

$\rho \cdot g$

$\times \text{Volume}$

Absolute weight of gas

Fluid mass and weight/

Eq. 3

Volume = 0.84 ft³

Compressed
air tank

Temp = 70°F, atm
pressure

14.7
psi

When tank filled with ~~air~~ ^($\frac{N}{m^3}$) gage pressure

50 psi density and weight of air!

$$PV = nRT \rightarrow \text{Mass specific form of ideal gas law} \quad \text{Pressure} = \frac{\text{number}}{V} \frac{RT}{\text{mole}}$$

from gage
Absolute Pressure

ideal gas law

$$P = P_c RT$$

$$P = \frac{P}{RT} = \frac{50 + 14.7}{(14.7) \frac{\text{ft} \cdot \text{lb}}{\text{slug} \cdot \text{R}}} \left(\frac{\text{lb}}{\text{in}^2} \right) \rightarrow \frac{(144 \text{ inch}^2)}{\text{ft}^2}$$

$$(14.7) \frac{\text{ft} \cdot \text{lb}}{\text{slug} \cdot \text{R}}$$

$$(70 + 460) \text{ R}$$

$$P = X + 460$$

Jerry
air compressor

Psi unit? : Pressure $\Rightarrow$ force / area

$$= \frac{\text{Pounds}}{\text{inch}^2} \left(\frac{\text{lb}}{\text{in}^2} \right)$$

R unit?

Gas
Constant.

$$R = \frac{P}{\rho \cdot T}$$

$$= \frac{1716 \text{ ft} \cdot \text{lb}}{\text{length force}}$$

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{\text{slug}}{\text{ft}^3}$$

$$\text{Pressure} = \frac{\text{Force}}{\text{area}} = \frac{\text{lb}}{\text{ft}^2}$$

$$\frac{\text{slug} \cdot R}{\text{weight}}$$

density of air

$$\rho = 64.1 \text{ lbf}$$

$$1716.530$$

$$= \frac{\text{ft} \cdot \text{lb}}{\text{slug} \cdot R}$$

$$\frac{\text{lb}}{\text{ft}^2}$$

$$\frac{\text{ft} \cdot \text{lb}}{\text{slug} \cdot R}$$

$$\text{slug} \cdot R$$

$$= \frac{\text{slug} \cdot \text{lb}}{\text{ft}^3 \cdot \text{lb}} =$$

$$\frac{0.012}{\#}$$

Absolute
value: pressure
temp

fluid mass and weight

Weight?

$$W = \text{specific}$$

$$\frac{\text{Force}}{\text{Volume}}$$

$$= \frac{\text{mass}}{\text{volume}} \cdot g = \rho \cdot g$$

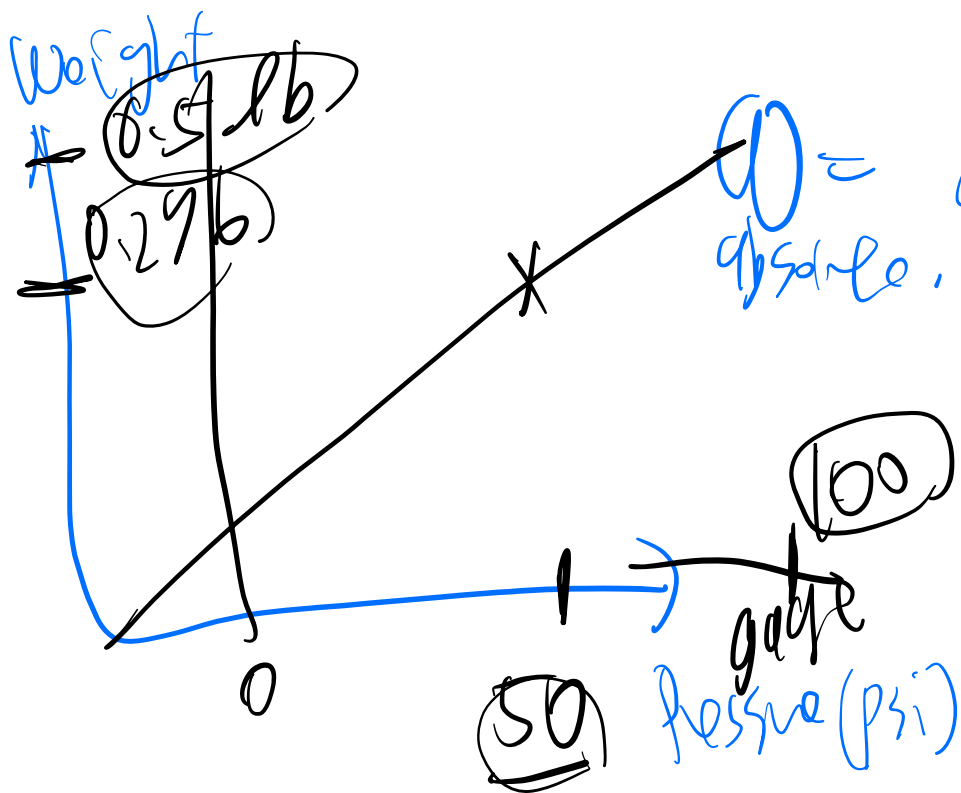
$$= 0.012$$

specific weight relative to air density.

$$W_{\text{total}} = \rho \cdot g \cdot \text{Volume}$$

$$0.012 \frac{\text{slug}}{\text{ft}^3} \cdot 32.2 \frac{\text{ft}}{\text{s}^2} \cdot 0.84 (\text{ft}^3) = 0.296 \frac{\text{slug} \cdot \text{ft}}{\text{s}^2}$$

$$1 \text{ pound} = \text{slug} \times \frac{\text{ft}}{\text{s}^2}$$



Eq. 4

Viscosity and dimensionless Quantities



length? D
pressure?

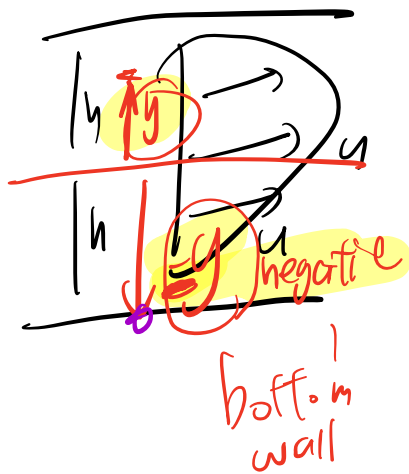
viscosity? speed?

$$\text{Viscosity} = 0.38 \frac{\text{N} \cdot \text{s}}{\text{m}^2}$$

$$\text{Viscosity} = \frac{\tau}{\frac{du}{dy}} = \frac{\text{shear stress } \frac{\text{Newton}}{\text{m}^2}}{\frac{\text{m}}{\text{s}}} = \frac{\text{N} \cdot \text{s}}{\text{m}^2}$$

Density = Specific Gravity $\times$ Den
Absolute

e.g.



Velocity u

$$u = \frac{3V}{2} \left[1 - \left(\frac{y}{h} \right)^2 \right]$$

Mean velo: $V = \frac{1}{2} \frac{du}{dy}$

$\mu_{\text{viscosity}} = 0.04 \frac{\text{lb} \cdot \text{s}}{\text{ft}^2}$

$\mu = \frac{\tau}{\frac{du}{dy}} = \frac{N}{m^2} \cdot \frac{s}{m} = \frac{N \cdot s}{m^2}$

Q
shearing stress
acting
on bottom wall,
acting on a
plane centerline,

$$\tau = \mu \cdot \frac{du}{dy}$$

$$\frac{du}{dy} = \left(-\frac{1}{h^2} \right) \cdot 2y \cdot \left(\frac{3V}{2} \right)$$

$= \left(\frac{3Vy}{h^2} \right)$ also along

the bottom wall $y = -h \rightarrow$ bottom wall, speed $\uparrow$ as y moves upward from $\frac{du}{dy} > 0$

On the bottom wall $\tau = \mu \cdot \frac{3V}{h}$

$\tau = 0$ midplane

$\tau = 14.4 \frac{\text{lb}}{\text{ft}^2}$

$\frac{du}{dy} = \left(\frac{3V}{h} \right)$ shear stress > 0

★ Think: Shear stress negative at upper wall! $\tau = -14.4$

$$\left(\frac{du}{dy}\right) = \frac{-3V}{h^2} \quad y = h \rightarrow \frac{du}{dy} = \frac{-3V}{h} \rightarrow \tau = \mu \cdot \frac{du}{dy} = -14.4$$

Shear stress: not like pressure

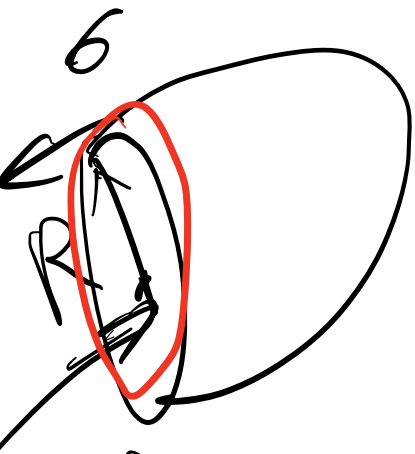
eq. 8 Capillary Rise in a Tube

Height $\rightarrow$
of
liquid in
vertical tube

Pressure Δp

Water 20°C

less than $h = 1.0$
mm



Force due to
Surface
Tension

$$= \underbrace{(2\pi R)}_{\text{unit length}} \cdot \sigma$$

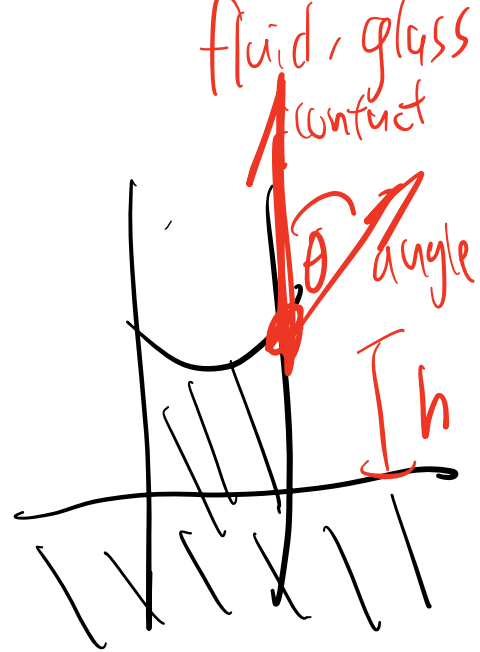
force on
half of droplet

$$= \Delta p (\pi R^2)$$

internal pressure p_i
 external pressure p_e

$$\Delta p = p_i - p_e = \frac{2\sigma}{R}$$

Surface tension: force per unit length!



Height govern by value of surface tension

σ , R , γ contact angle
weight of liquid

$$\gamma = \rho \cdot g$$



Vertical force due to surface tension

$$= (2\pi R) \cdot \sigma \cdot \cos \theta$$

$$= \text{Weight} = \underbrace{\gamma \cdot \pi R^2 \cdot h}_{\text{Volume}}$$

Practice
1.85

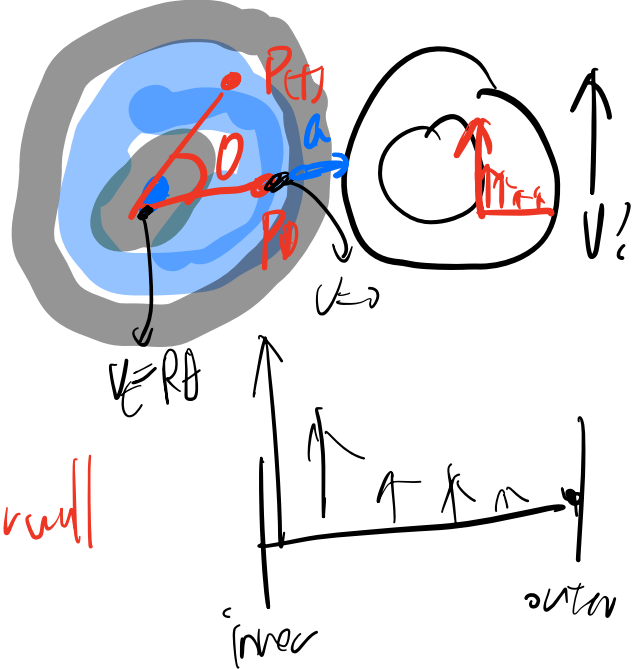
viscosity

fixed

glycerin

inner wall

outer wall



angular
velocity
 ω

$$\frac{6\pi R\omega}{5}$$

How
fast

it spins
around center point

$$\omega = \frac{\Delta\theta}{\Delta t}$$

for a full
rotation

$$2\pi R$$

$$V = \frac{\text{distance}}{\text{time}}$$

$$= \frac{S}{\text{time}} \quad (S = \text{arc length})$$

$$S = R\theta$$

$$\frac{ds}{dt} = V = R \cdot \frac{d\theta}{dt} = R\omega$$

shear
stress
outer - inner

$$\tau = \mu \cdot \frac{du}{dy}$$

Weight per volume - Specific weight Force: Newton - $\text{kg} \cdot \text{m/s}^2$ ^{thermo} value in Mass (Volume)
1) Fluid Character, dimension, unit, Density/Temp

Analysis of fluid behavior (Measure of fluid mass ^{mass per unit for fluid} weight)
1st and 2nd thermodynamics No slip condition density kg/m^3
Ideal gas law, Viscosity, who fluid has viscosity Specific weight, gravity
shear stress Compressibility of fluids,

Vapor pressure, Surface tension, bulk modulus
Contact angle wetting, capillary effect Compression and Expansion of gases, speed of sound.

2) Fluid statics, pressure at a point

Pressure Field, Pressure variation in a fluid at Rest, Incompressible, Compressible fluid, stand and atomosphere

Measurement
of
pressure

manometry
Piezometer
U-tube Manometer
Inclined-Tube manometer

Hydrostatic
force on
plane
surface

Pressure
Prism

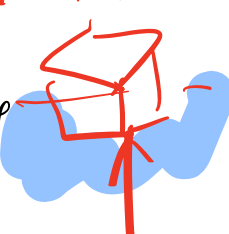
hydrostatic
Force on
plane surface

Pressure
prism
(3D-model)

Device
Mechanical
Electronic
Pressure Measure

* Hydrostatic
force
on
curved
Surface

* Buoyancy
Flotation
Stability
by liquid to object
upward force



Pressure variation in a fluid
with rigid body motion

Linear
motion
rigid body rotation

3) The Bernoulli Equation

fluid dynamics

Newton's
Second law
 $F = ma$

along streamline
Normal to
streamline

physics
inter
alternative
form

Static, stagnation
Dynamic and
total pressure

Example Bernoulli
of use

Free jets
Confined flows
Flow rate measurement

Restriction on
the use of
the Bernoulli

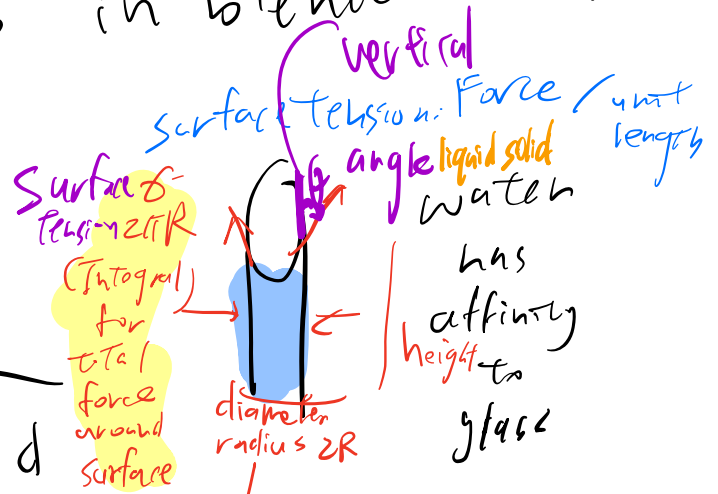
compressibility effects
unsteady effects
rotational effects

9/26/20 lecture Fluid Statics

Microfluidics Channel:

Cavitation bubbles in blender - vid.

Capillary Effect



a liquid-gas-solid

interface we may have

adhesion of liquid to solid.

Draw 3 body diagram
 $h = \frac{\sigma \cos \theta}{\rho g R}$
 mass per unit volume
 water is held in tube around the perimeter

One consequence is fluid pulling this up a

tube

$\sum F_y = 0$ Force is x canceled each others!

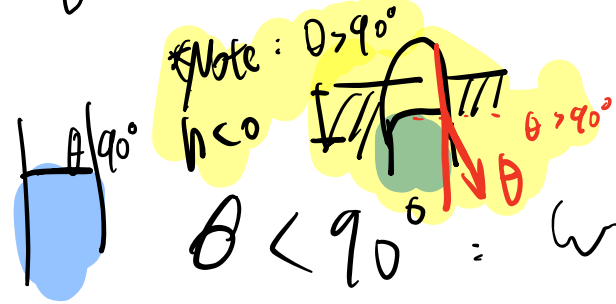
$$\cos \theta 2\pi R \sigma = \rho g \pi R^2 h$$

$$h = \frac{2\sigma \cos \theta}{\rho g R} = \frac{2\sigma}{\rho g R} \cos \theta$$

Capillary Effect
 $R \rightarrow h$

A: I don't necessary know, surface tension
 water-air
 can be look up

What if we change glass to metal
other solid...?



$\theta < 90^\circ$: wetting liquid

non wetting liquid? Mercury

wetting angle

$\theta > 90^\circ$: Non wetting liquid

$\rightarrow \cos \theta$ negative

σ property liquid-gas interface

θ

"

liquid-solid interface

Surface

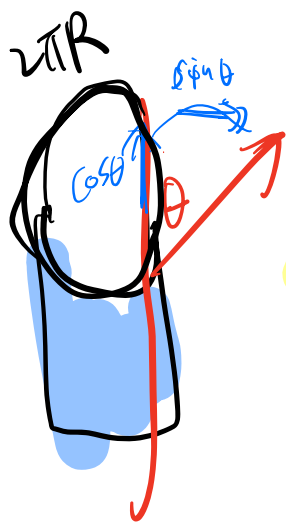
Non wetting

water-repellent



glass tube

coated wax tube



water-glass

σ = surface tension

$$F_x = \frac{\sigma}{2\pi R} = \frac{\text{force}}{\text{length over which force acts}}$$

surface
Upward force by surface tension

eg. Consider 50 μm glass eg
tube, wetting angle ≈ 0 Diagnostic Blood Microfluidics

Q surface tension!

$$\sigma = 0.0734$$

$$h = \frac{2(0.0734 \frac{\text{N}}{\text{m}}) \cos(0)}{1000 \frac{\text{kg}}{\text{m}^3} \cdot 9.81 \frac{\text{m}}{\text{s}^2}}$$

$$25 \times 10^{-6} \text{ m}$$

$$= 0.6 \text{ m} = 60 \text{ cm}$$

$$\frac{\text{kg} \cdot \text{m}}{\text{s}^2} \cdot \frac{\text{N}}{\text{m}} = \frac{\text{kg}}{\text{m} \cdot \text{s}^2}$$

Newton / meter

$$= \frac{\text{m}}{\text{unit}} \#$$

Newton:
mass $\cdot$ acceleration
 $= \text{kg} \cdot (\text{m}/\text{s}^2)$

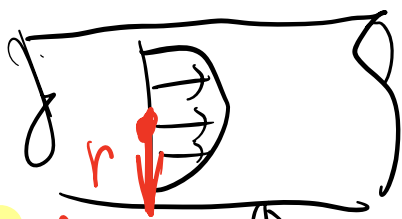
e.g. Consider designing
 vascular implant that coats
 inside of an artery and
 release some drug.

fluid statics

in viscous flow

superficial artery insert
catheter.

Suppose: $u(r) = u_{max} \times \left(1 - \frac{r^2}{R^2}\right)$



$u_{max} = 1 \text{ m/s}$

viscosity of blood?

$\mu = 0.004 \frac{\text{N}\cdot\text{s}}{\text{m}^2}$

$L = 8 \text{ cm}$

$R = 5 \text{ mm}$

$\tau = \mu \cdot \frac{du}{dy}$

velocity m/s

meters

integrate 0 ~ L over that surface

dr : from middle to the wall

$\geq R$: artery radius

fluid is exerting shear stress

Shear stress is not static, being max on the wall, min in the middle / different of velocity between layers of fluid.

$$\tau_w = \mu \cdot \frac{du}{dy}$$

Orthogonal to the flow direction for cylindrical coordinate



direction of blood flow

also u is a function of y

Shear Flow velocity Profile



$$\tau = \mu \frac{du}{dy}$$

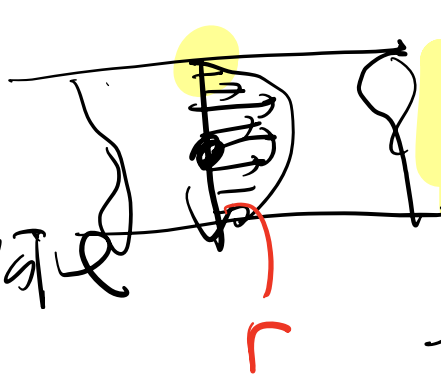
why is shear stress

positive?

now

$$\frac{du}{dy}$$

negative



Shear stress

$$\tau = \mu \frac{du}{dr}$$

at the wall

$r=R$
?

$r=0$ at center

$r=R$ at wall

Velocity Gradient at the wall

for parabolic profile $\rightarrow$ velocity $u(r) = U_{max} \left(1 - \frac{r^2}{R^2}\right)$

$$\frac{du}{dr} = - \frac{2U_{max}r}{R^2}$$

At the wall

$$\tau_w = -u \frac{du}{dr} \bigg|_{r=R} = \left(\frac{-2U_{max}}{R} \right) \bigg|_R$$

shear stress

$0.004 \frac{N}{m^2} \cdot s$

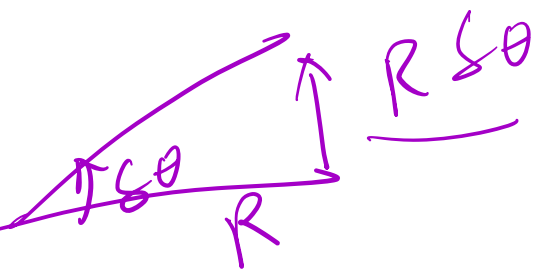
$\left(\frac{-2}{0.005m} \right)$

$$= 1.6 \frac{N}{m^2}$$

$$D = \int_0^{2\pi} \int_0^L \tau_w R dz d\theta$$

$L = 8cm$

$R = 5mm$



integrating cylindrical
arc length $R d\theta$ why?

$$= \tau_w \cdot R \cdot (L) \cdot (2\pi)$$

$$= 1.6 \frac{N}{m^2}$$

Question: Torque!

Hint: Bottom of cylinder
function of r

Integral

